

RESEARCH REPORT
DATASET COLLECTION REPORT
Trinetra Drones and Robotics Pvt. Ltd.



Submitted By
Anshika Kodukula
SRM UNIVERSITY-AP
AP24110020029

Dataset Collection Report for Crop Disease Detection and Crop Health Monitoring

Introduction

Artificial Intelligence models require high-quality datasets for training and testing. Agricultural datasets containing crop images are essential for developing systems that can identify diseases, monitor crop health, and support precision farming. This report presents a collection of publicly available datasets used in agricultural AI applications.

Dataset 1: PlantVillage Dataset

PlantVillage is one of the most widely used agricultural image datasets for crop disease detection.

Features

- More than 50,000 images
- Multiple crop species
- Healthy and diseased plant leaves
- Used extensively in machine learning research

Applications

- Disease classification
- Crop health monitoring
- Deep learning model training

Dataset 2: Rice Leaf Disease Dataset

This dataset contains images of rice leaves affected by various diseases.

Features

- Rice crop images
- Multiple disease categories
- Suitable for disease detection research

Applications

- Rice disease identification
- Automated crop monitoring

Dataset 3: Tomato Leaf Disease Dataset

The dataset includes images of healthy and diseased tomato leaves.

Features

- Multiple disease classes
- High-quality leaf images
- Commonly available through Kaggle

Applications

- Tomato disease detection
- Deep learning model development

Dataset 4: Corn Leaf Disease Dataset

This dataset focuses on disease detection in corn crops.

Features

- Healthy and infected leaf images
- Multiple disease categories
- Useful for image classification tasks

Applications

- Corn disease diagnosis

- Crop health assessment

Dataset Comparison:

Dataset Name	Crop Type	Application
PlantVillage	Multiple Crops	Disease Detection
Rice Leaf Disease Dataset	Rice	Disease Classification
Tomato Leaf Disease Dataset	Tomato	Disease Detection
Corn Leaf Disease Dataset	Corn	Crop Health Monitoring

Importance of Agricultural Datasets

Agricultural datasets help researchers and developers:

- Train AI models
- Improve disease detection accuracy
- Monitor crop health
- Develop smart farming solutions

Conclusion

Publicly available agricultural datasets provide valuable resources for developing AI-based crop monitoring and disease detection systems. Among the identified datasets, PlantVillage is the most comprehensive and widely used for agricultural research and machine learning applications.